



GIT REPOSITORY

A distributed version control system





Outline

- Introduction to source control
 - What is version control system
 - History of source control
 - Centralized vs Distributed version control
- Introduction to Git
 - What is Git
 - Git Pros and Cons
 - Git workflow
 - Git Architecture
 - Git at work





WHAT IS VERSION CONTROL SYSTEM?

- a way to manage files and directories
- track changes over time
- recall previous versions
- 'source control' is a subset of VCS



HISTORY OF SOURCE CONTROL

- (1972) Source Code Control System (SCCS)
 - - closed source, part of UNIX
- (1982) Revision Control Systems(RCS)
 - - open source
- (1986) Concurrent Version System(CVS)
 - - open source



Source Code Control System
Proprietary software



Revision Control System
GNU General Public License



Concurrent Versions System
GNU General Public License



HISTORY OF SOURCE CONTROL

- (2000) Apache Subversion (SVN)
 - - open source
-
- (2005) Git
 - - open source
 - - distributed version control system



Apache Subversion
Apache License



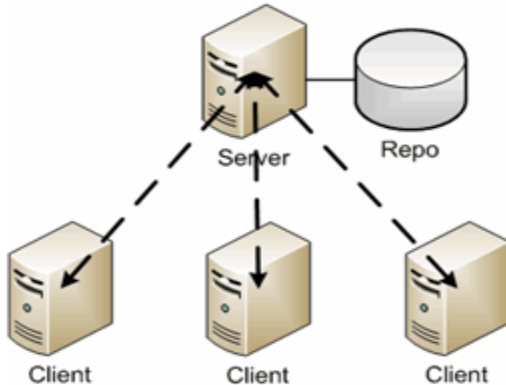
Git
GNU General Public License



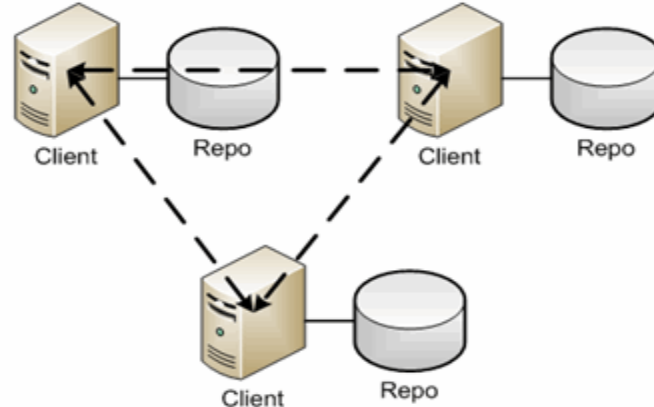
CENTRALIZED VS DISTRIBUTED VERSION CONTROL

- no central server
- every developer is a client, the server and repository

Traditional



Distributed





WHAT IS GIT?

- created by Linus Torvalds, April 2005
- replacement for BitKeeper to manage Linux kernel changes
- a command line version control program
- uses checksums to ensure data integrity
- distributed version control
- cross-platform(including windows)
- open source, free





GIT PROS AND CONS

PROS

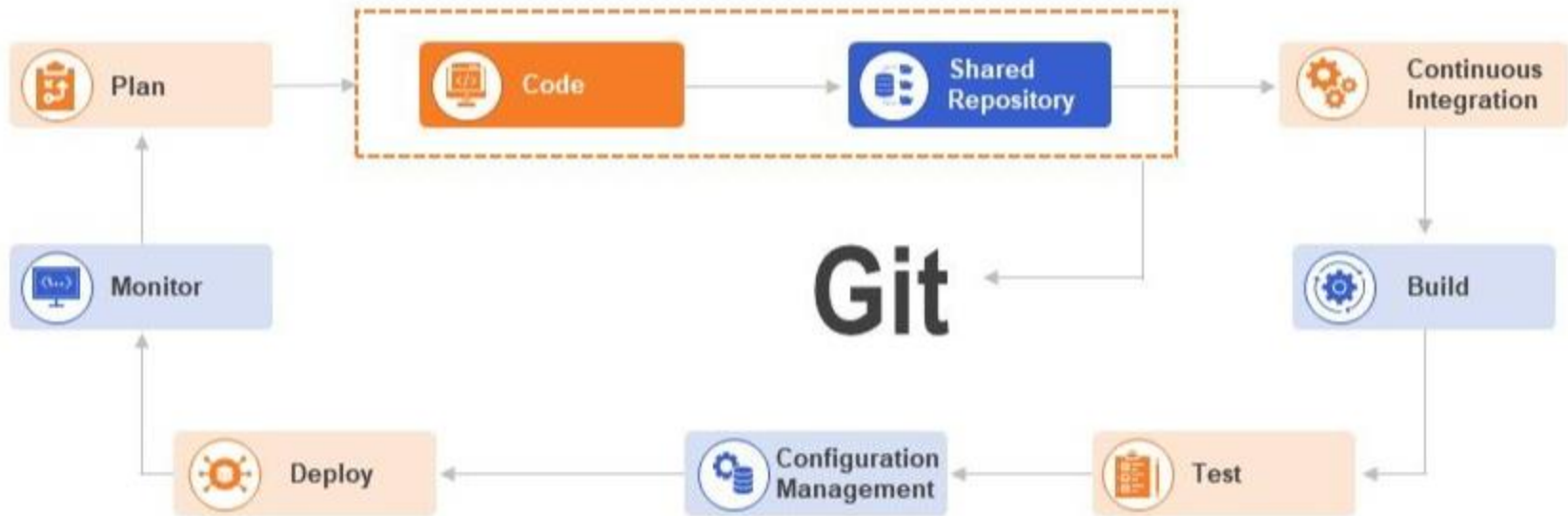
- Operates locally
- Avoids having a single point of failure.
- Handles merging from multiple contributors effectively.

CONS

- Has a higher learning curve
- Lacks granular access control
- Does not effectively handle storing large binary files.

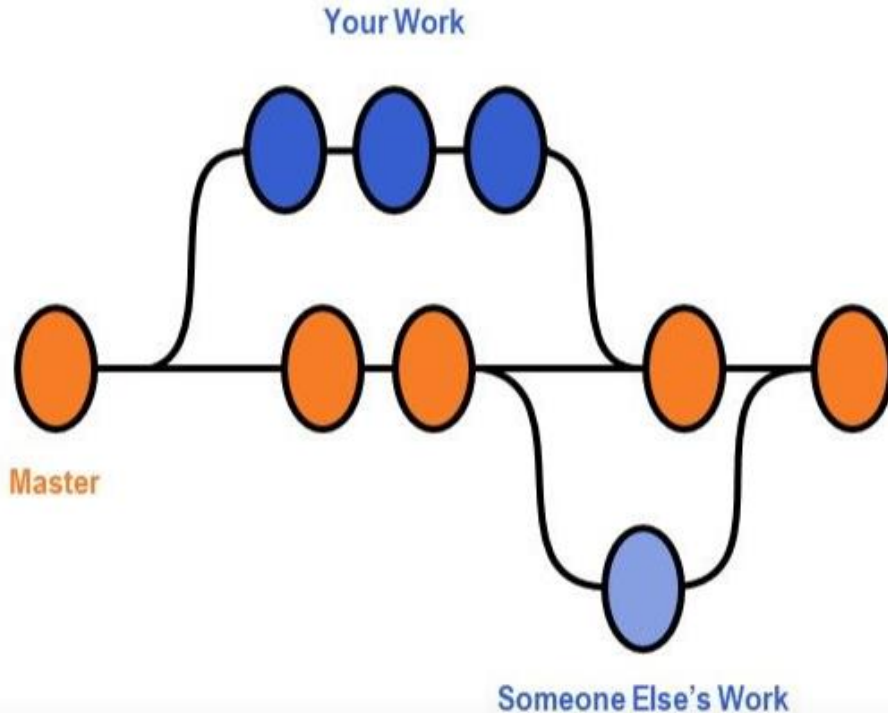


GIT WORKFLOW





GIT WORKFLOW



- > Git is the foremost ordinarily used version system
- > Git tracks the changes you create to files, therefore you've got a record of what has been done, and you'll revert to specific versions do you have to ever need to
- > Git also makes collaboration easier, permitting changes by multiple folks to all or any be incorporate into one supply



01. Performance

Git performs terribly powerfully and faithfully compared to alternative version management systems. New code changes is simply committed, version branches is effortlessly compared and incorporated, and code can even be optimized to perform higher. Algorithms employed in developing stinkpot take the complete advantage of the deep information hold on inside, with regards to the attributes accustomed produce real ASCII text file file trees, however files area unit changed over time and what quite file access patters area unit accustomed recall code files as and once required by developers. stinkpot primarily focus upon the file content itself instead of file names whereas determinative the storage and file version history. Object formats of stinkpot repository files use many mixtures of delta coding and compression techniques to store data objects and directory contents.



02. Security

Git is intended specially to keep up the integrity of ASCII text file. File contents moreover because the relationship between file and directories, tags, commits, versions etc. area unit secured cryptographically mistreatment associate formula referred to as SHA1 that protects the code and alter history against accidental moreover as malicious harm. you'll be able to make sure to own associate authentic content history for your source code with stinkpot.



03. Flexibility

A key style objective of stinkpot is that the quite flexibility it offers to support many sorts of nonlinear development workflows and its potency in handling each tiny scale and enormous scale comes moreover as protocols. it's unambiguously designed to support tagging associated branching operations and store every and each activity meted out by the user as an integral a part of "change" history. Not all VCSs support this feature.



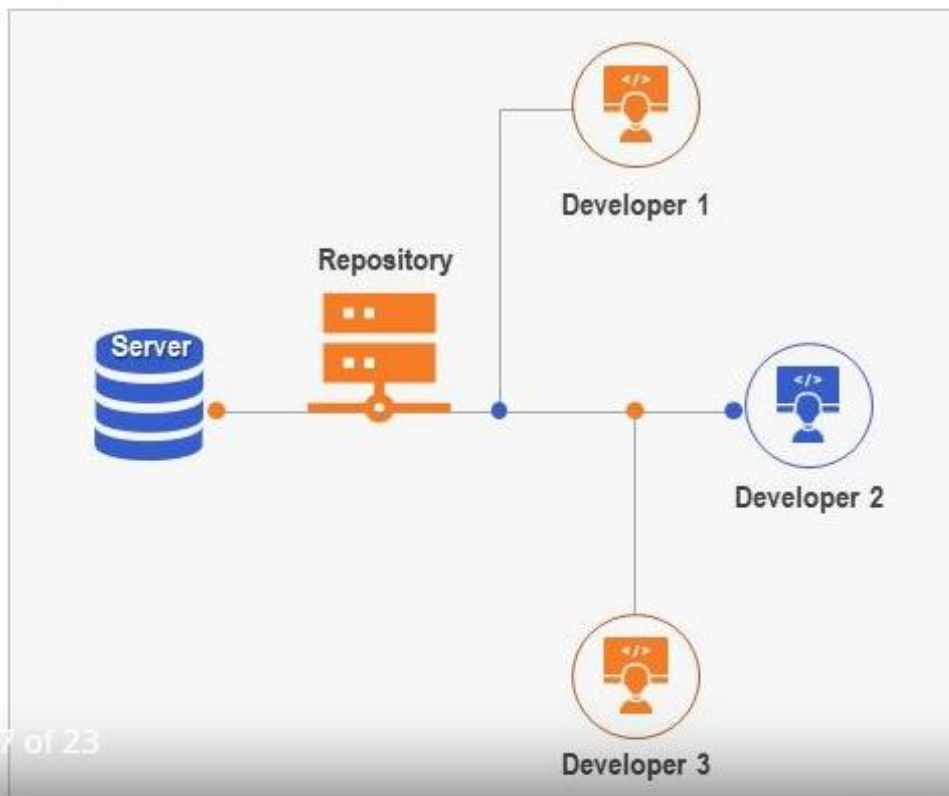
04. Wide Acceptance

Git offers the kind of performance, practicality, security and suppleness that almost all developers and groups have to be compelled to develop their comes. compared to alternative VCS stinkpot is most generally accepted system because of its universally accepted usability and performance standards.



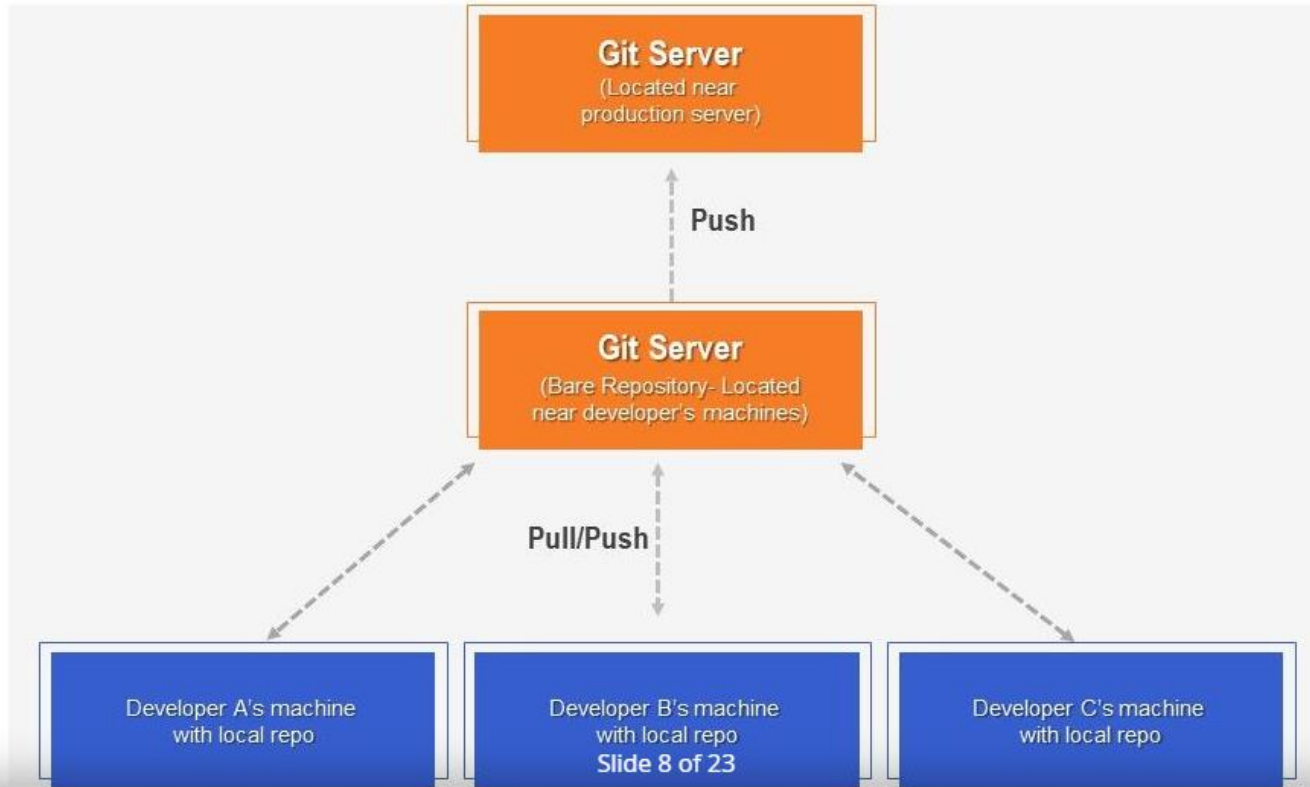
05. Quality Open Supply Project

Git may be a wide supported open supply project with over 10 years of operational history. individuals maintaining the project area unit all right matured and possess an extended term vision to fulfill the future wants of users by cathartic staged upgrades at regular intervals of your time to boost practicality moreover as usability. Quality of open supply software system created offered on stinkpot is heavily scrutinized a multitudinous variety of times as businesses nowadays rely heavily on stinkpot code quality.



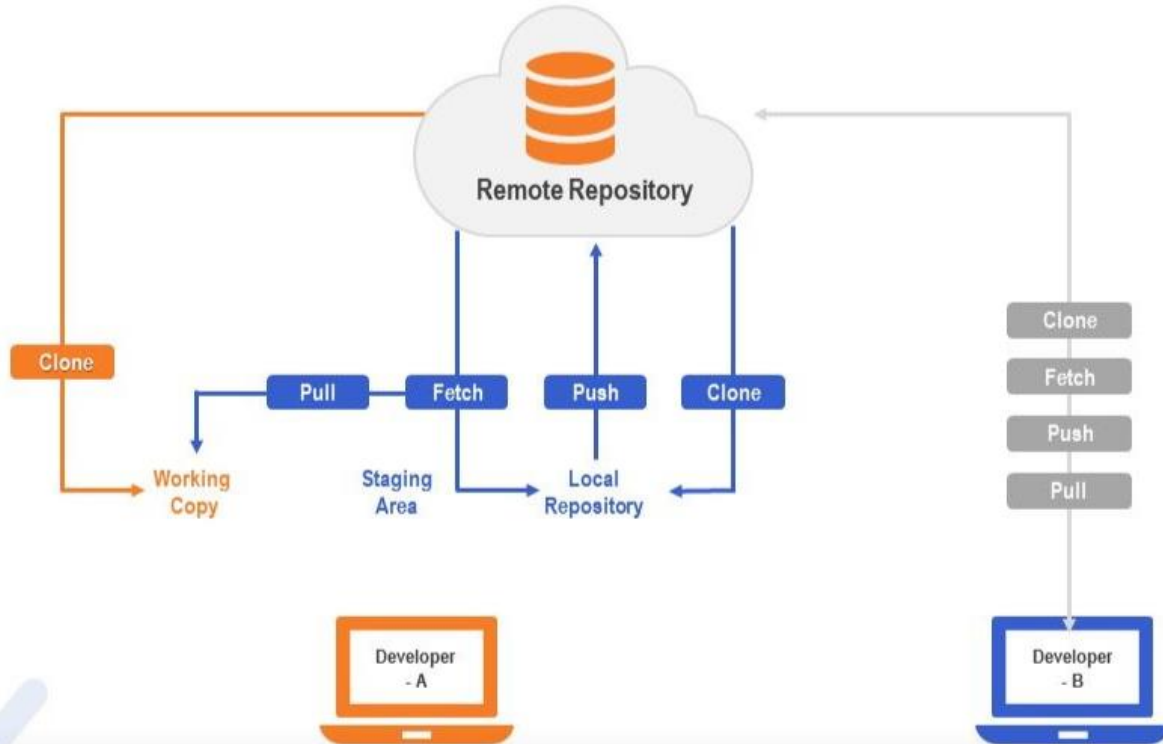


ARCHITECTURE OF GIT





GIT AT WORK





STEPS HOW TO USE GIT

- Create Git account in github.com
- Click your profile from the upper right corner
- Then Click Repositories
- Look for NEW Repositories Button then Click
- Create New Repository
- Fill in repository name
- Then click Create Repository
- Copy the url to clone
- Then use from the next step in cloning
- Git clone URL HERE



STEPS HOW TO USE GIT

- Download and install git bash <https://git-scm.com/downloads> or you can use powershell, terminal for mac or any alike application
- Open any terminal installed
- Create folder on your local station folder (e.g xampp/htdocs/foldername)
- Git clone <urlofgitrepository>
- Locate your newly cloned files then type **ls** to check files.
- Type git status – all files in red means needs to push to repository
- Type git status – all in red files must be gone – all files must be in green
- Type git add * - * means all in red files that you want to commit
- Type git commit –m 'reason of commit'
- Type git status
- Type git pull
- Type git push



STEPS HOW TO ADD COLLABORATOR

- Go to your repository
- Click the repository name
- Click Settings
- Click collaborators button
- Then fill in my account name **wizgabriel18**



THANK YOU